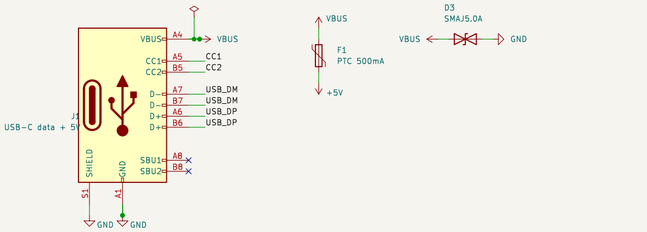
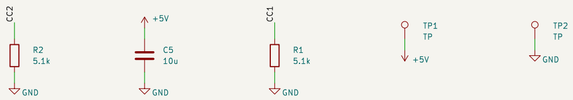
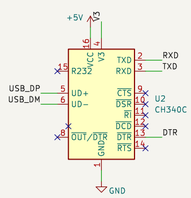
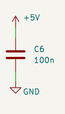
USB-C INPUT: data pair to CH340C, CC pull-downs, PTC, TVS





USB-SERIAL: CH340C, auto-reset capacitor

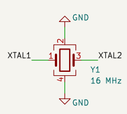






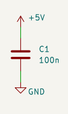


MCU: ATmega328P-AU, 16 MHz crystal





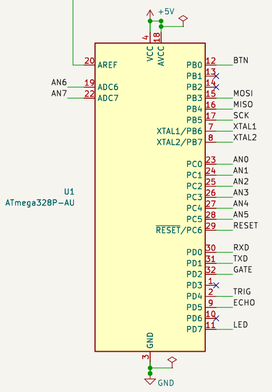




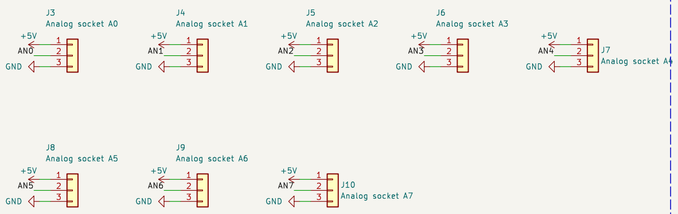






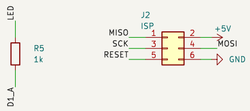


8 ANALOG SENSOR SOCKETS A0–A7



PHOTOGATE, ULTRASONIC, REACTION PAD, PROGRAMMING





Prayog hub: Arduino-Nano-class board. CH340C gives USB serial and auto-reset (DTR through 100 nF). Sockets are +5V / signal / GND JST-PH; photogate on D2 (interrupt), ultrasonic trigger D4 and echo D5, reaction pad button D8 and LED D7. Flash firmware/release/prayog.hex over USB with the Arduino bootloader or over ISP.

First engineering release; see pcb/README.md for status and limits

ClaudeWork

Sheet: /
File: prayog_hub.kicad_sch

Title: Prayog STEM lab kit — hub board

Size: A2	Date: 2026-11-01	**Rev: A**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1